

Question	Marking Guidance	Mark	Comments
04.1	Substitution;	1	Accept inversion or translocation Ignore 'point mutation'
04.2	1. ($\text{VO}_2 \text{ max}$ and CS activity) increased for both groups; 2. No statistical test, so do not know if differences are significant OR No statistical test, so differences could be due to chance; 3. Only 8 weeks training OR Training did not last long; 4. Might not be true for all types of training/exercise/females;	3 max	Max 2 marks for mark points 2, 3 and 4 2. Ignore standard deviation 2. Accept correct named statistical test eg t-test
04.3	In Group C : 1. Less mitochondrial replication/production; 2. Less transcription (of genes) for mitochondrial proteins/CS OR Less translation of (mRNA into) mitochondrial proteins;	2	1. and 2. Accept converse for Group T 2. Accept less CS/enzyme is produced

04.4	For (no mark) 1. (From Figure 5 Group T have) increased CS activity for Krebs cycle; 2. (from Figure 4 Group T have) increased $VO_{2\text{ max}}$ so more oxygen (available) OR (from Figure 4 Group T have) increased $VO_{2\text{ max}}$ so more aerobic respiration OR (from Figure 4 Group T have) increased $VO_{2\text{ max}}$ so delayed anaerobic respiration; Against (no mark) 3. No correlation between (percentage change in) $VO_{2\text{ max}}$ and CS activity OR No correlation on Figure 6; 4. It might not be thymine causing the increase OR There may be other differences in the control region (of Group T) that cause the increase; 5. $VO_{2\text{ max}}$/CS activity not the only measures of ability to exercise for longer;	3 max	Max 2 marks for mark points 3, 4 and 5 Ignore any answers relating to sample size or duration of investigation Ignore 'correlation does not mean causation' unless qualified 2. Accept 'less lactate' for delayed anaerobic respiration 4. Ignore 'could be due to lifestyle/diet changes' 5. Accept ideas that they did not measure ability to exercise for longer
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